Unit 5

Gravitational Fields

Unit 5 : Chapter One

Gravitational Force Field

1.1 Gravitational Fields

During the initial decades of the 13th century, the scientist Keplar, having done investigations into the motion of celestial bodies has recorded, his observations about the elliptical motion and the periods of revolution of these bodies.

A young scientist Isac Newton who lived during the mid era of the same century carrying investigations on the approximate circular motion of the planets in the solar system, around the Sun revealed that a certain force of attraction made by the sun on them helps in their circular motions.

Furthermore, he confirmed that it is this same force of attraction that holds the moon in its orbit and also brings the apple on to the Earth leaving the tree.

On further research it was proved that in the universe, between any two particles having masses exist these mutual forces of attraction mutually and these forces were named as "gravitational forces".

Broad research was subsequently done by Newton on factors affecting this force of attraction and he, being a mathematician too, expressed his findings mathematically by the following law.

1.2 Newton's Law of Gravitation

The gravitational force of attraction between two particles of masses M and m situated at a distance r is given by,

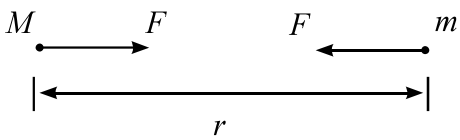


Figure 1.1

$$F = G\frac{Mm}{r^2}$$

where G is a universal constant known as Gravitational constant. ($G = 6.67 \times 10^{-11}$ N m^2 kg^{-2})

The gravitational field has been introduced as a new concept for gravitational force. According to this concept, there exists a gravitational field around any mass. The gravitational force of attraction between two masses is considered as the result of the inter-action between the two fields when they intersect. Hence, a gravitational field is a force field.

1.3 Intensity of a Gravitational field

The strength of a gravitational field at any point in it is measured by the quantity known as the " Gravitational field intensity " at that point. The intensity of a gravitational field at such a point is the force acting on a unit mass placed at that point.

Hence, if F is the force acting on a mass m placed at a certain point in a gravitational field, the gravitational field intensity at the point is,

$$g = \frac{F}{m} \text{ N kg}^{-1}$$

The gravitational field due to an isolated mass can be mentioned as the simplest of all gravitational fields.

1.3.1 Gravitational field intensity at a point in the gravitational field due to an isolated mass

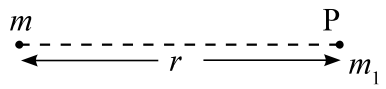


Figure 1.2

P is a point in the gravitational field due to the isolated mass m, situated at a distance r from the same mass. In order to determine the gravitational field at P let us place a mass element Δm at P and find the force on it.

According to Newton's law of gravitation,

The force mass m_1, $(F) = G \dfrac{m\, m_1}{r^2}$

Hence, gravitational field intensity $g = \dfrac{F}{m_1} = G\dfrac{m}{r^2}$

Gravitation is essentially a force of attraction. Hence the above intensity is a vector quantity directed towards the mass m.

The variation of gravitational intensity with the distance from the isolated point mass can be represented graphically as follows.

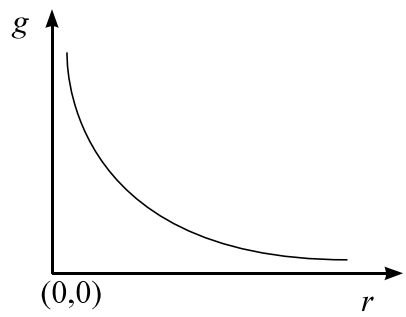


Figure 1.3

1.3.2 Gravitational Intensity around a spherical mass

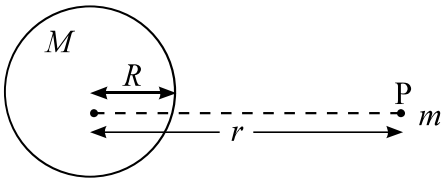


Figure 1.4

P is a point situated at a distance r ($r > R$) of a spherical body of mass M and radius R. In order to determine the gravitational intensity at P, let us place a mass element m at P and find the force on it.

Assuming the whole mass of the sphere is accumulated at its centre.

$$F = G\frac{Mm}{r^2}$$

$\therefore$ gravitational field intensity at P, $(g) = \dfrac{F}{m} = G\dfrac{M}{r^2}$

Similarly gravitational field intensity on the surface of the sphere, $g = \dfrac{GM}{R^2}$

1.4 Gravitational potential

Since any mass placed at a certain point in a gravitational field is subjected to a gravitational force, a certain amount of work is done by the field in bringing this mass from outside the field (infinity) to this point. While this work is accumulated as potential energy and the work done per unit mass in bringing it to this point is referred to as the " **gravitational potential**" at the point.

1.4.1 Gravitational potential at a point in the gravitational field due to an isolated mass

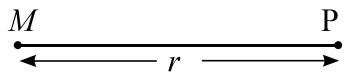


Figure 1.5

P is a point in the gravitational field due to mass M, and is situated at a distance r from the same mass. It is possible to derive mathematically that the work done in bringing unit mass from infinity up to the point P is, $W = -G\dfrac{M}{r}$

$\therefore$ gravitational potential at point P, $= V = -G\dfrac{M}{r}$

The negative sign above indicates that the relevant work is done by the field and not against the field. Also since work is a scalar, gravitational (field) potential too is a scalar quantity.

The following graph illustrates the variation of gravitational potential with the distance from a point mass.

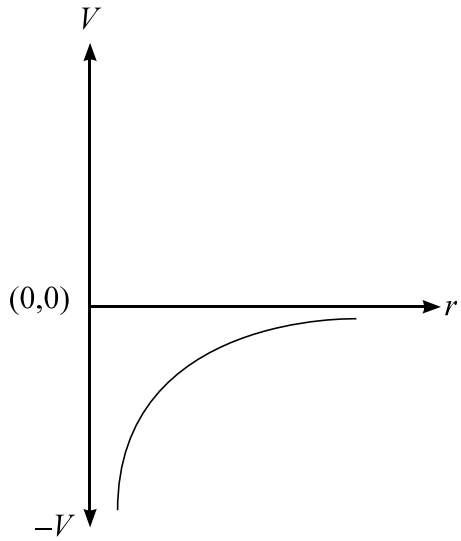


Figure 1.6

1.4.2 Potential energy of a mass placed in a gravitational field

If V is the potential at a point in a gravitational field, the potential energy of a unit mass placed at that point too is V.

$\therefore$ Potential energy of a mass placed at the point, $E = mV$

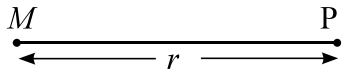


Figure 1.7

If the above point is situated in the gravitational field due to a mass M, at a distance r from M, then the gravitational potential at point P,

$$V = -G\frac{M}{r}$$

$\therefore$ The potential energy of a mass placed at the point P, $E = mV = -G\dfrac{Mm}{r}$

Unit 5 : Chapter Two

Earth's Gravitational Field

According to the mass of the Earth, a specific gravitational field has been created around it.

2.1 Gravitational field due to the Earth

Considering Earth as a spherical body of mass M and radius R, the gravitational force of a body of mass m placed on the surface of Earth,

$$F = G\frac{Mm}{R^2}$$

Also the force on a mass m placed on the surface of Earth is its weight.

ie: $F = mg$, where g = acceleration due to gravity

$$\therefore \quad mg = G\frac{Mm}{R^2}$$

Therefore, gravitational field intensity can be written as,

$$g = G\frac{M}{R^2}$$

Hence it is seen that the acceleration due to gravity and the gravitational field intensity on the surface of Earth take the same value.

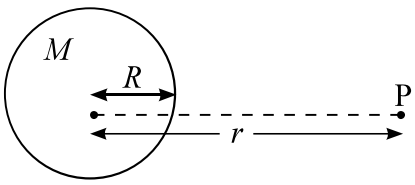


Figure 2.1

Consider a mass m placed at a point P at a distance r ($> R$) from the centre of the Earth.

The gravitational force on mass m, $F = G\frac{Mm}{r^2}$

$\therefore$ Gravitational field intensity, $g = \frac{F}{m} = G\frac{M}{r^2}$

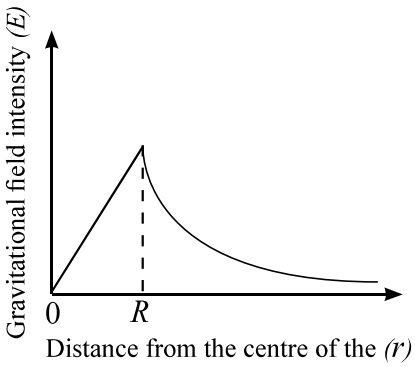


Figure 2.2

The variation of the above intensity at points from the centre of Earth to the surface and outside Earth's surface, with the distance from the centre of Earth can be shown graphically as given in Figure 2.2.

2.2 Potential energy at a height h in the gravitational field of the Earth

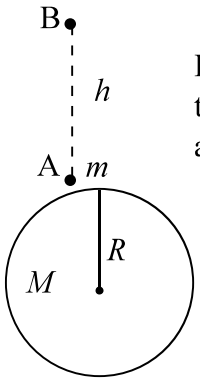


Figure 2.3

Figure 2.3 shows a body of mass 'm' being raised from a point A close to the surface of Earth to a point B at a height h. Let M be the mass of Earth and R its radius.

Gravitational potential at point A $= -G\dfrac{M}{R}$

Potential energy possessed by the body at point A $= -G\dfrac{Mm}{R}$

Gravitational potential at point B $= \dfrac{-GM}{(R+h)}$

Potential energy possessed by the body at point B $= \dfrac{-GMm}{(R+h)}$

If E is the excess potential energy received by the body when raised from A to B,

$$E = \frac{-GMm}{(R+h)} - \left(\frac{-GMm}{R}\right)$$

$$E = \left[\frac{-GMm\,h}{(R+h)\,R}\right]$$

When $h << R$, $(R + h)\,R \simeq R^2$

$\therefore$ Hence $E = \dfrac{GMm\,h}{R^2}$ ——— (1)

If g is the gravitational acceleration,

$$g = \frac{GM}{R^2} \quad \text{——— (2)}$$

From the above two equations it can be derived that $E = mg\,h$

Hence the gravitational potential energy possessed by a body relative to a zero energy level is given as $E = mgh$ where h is the height of the body above the zero level.

Note:
This expression can be used only for very small values of h compared with R.

2.3 Motion of a satellite around Earth.

Many satellites revolve around our Earth. Out of these satellite the only natural satellite is the Moon. The motion of the Moon as well as of all other artificial satellites is based on common principles.

For example, consider a satellite of mass m revolving around the Earth in an orbit of radius *r* with a speed *v*,

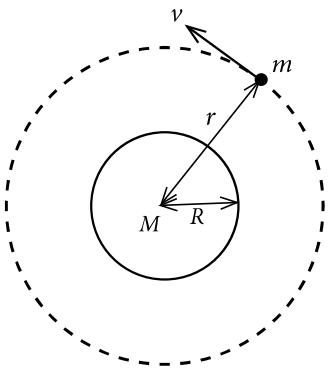


Figure 2.5

The centripetal force $\left(\frac{mv^2}{r}\right)$ required for this circular motion is provided by the gravitational force of attraction of the Earth.

If *M* is the mass of the Earth, then,
According to the equation $F = ma$,
[$\left(\frac{v^2}{r}\right)$ = centripetal force]

$$\left(G\frac{Mm}{r^2}\right) = \left(\frac{mv^2}{r}\right)$$

$$\therefore \quad v^2 = \frac{GM}{r}$$

Hence, speed of the satellite,

$$v = \sqrt{\frac{GM}{r}}$$

The periodic time of revolution of the satellite *(T)*,

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{v/r} = \frac{2\pi r}{v} = \frac{2\pi r}{\sqrt{\frac{GM}{r}}}$$

$$T = 2\pi\sqrt{\frac{r^3}{GM}}$$

The quantities *G* and *M* are constants. Hence from the above expression it is seen that the speed of a satellite and its period of revolution depend only on the radius of its orbit.

When a satellite revolving close to the Earth is considered,
Radius of the orbit *(r)* ≈ Radius of the Earth *(R)*

Also if '*g*' is the acceleration due to gravity on the Earth, the weight of any body on it is equal to the gravitational force of attraction.

$$mg = G\frac{Mm}{R^2}$$

$$\therefore GM = gR^2$$

By substituting the above expressions,

Speed, $v = \sqrt{\frac{GM}{r}} = \sqrt{\frac{gR^2}{R}} = \sqrt{gR}$

Periodic time of revolution, $T = 2\pi\sqrt{\frac{R^3}{gR^2}} = 2\pi\sqrt{\frac{R}{g}}$

2.3.1 Calculation of the speed and periodic time of a satellite revolving close to Earth

Let us consider the radius of the Earth $R = 6.4 \times 10^6$ m and $g = 10$ m s^{-2}
By substituting to the equation $v = \sqrt{gR}$

Satellite speed $v = \sqrt{10 \times 6.4 \times 10^6} = \sqrt{64 \times 10^6} = 8 \times 10^3$ m s^{-1}

By substituting to the equation $T = 2\pi\sqrt{\frac{R}{g}}$,

Periodic time , $T = 2\pi\sqrt{\frac{6.4 \times 10^6}{10}} = 5028$ s

≈ 1 h 24 min

2.3.2 Energy of a revolving satellite

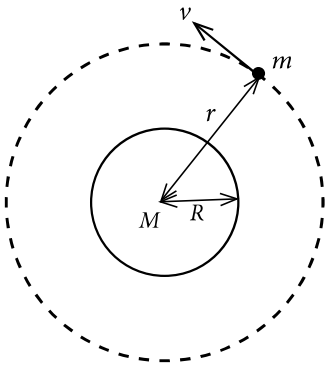


Figure 2.6

Consider a satellite of mass *'m'* moving in an orbit of radius *'r'* at a speed *'v'* around Earth of mass *M* radius *R*.

Then the gravitational potential at any point on the orbit, $V = -G\frac{M}{r}$

$\therefore$ Gravitational potential energy of the satellite moving along the orbit, $(E_1) = -G\frac{Mm}{r}$

Also, the kinetic energy of the satellite $(E_2) = \frac{1}{2}mv^2$.

But, the centripetal force = gravitational force

$$\frac{mv^2}{r} = G\frac{Mm}{r^2}$$

$$\therefore \quad \frac{1}{2}mv^2 = \frac{1}{2}G\frac{Mm}{r}$$

$$E_2 = \frac{1}{2}G\frac{Mm}{r}$$

$\therefore$ Total energy of the satellite, $E = E_1 + E_2 = -G\frac{Mm}{r} + \frac{1}{2}G\frac{Mm}{r}$

$$E = -\frac{1}{2}G\frac{Mm}{r}$$

The above expression is known as the energy equation of a satellite.

2.3.3 Geo - stationary satellites

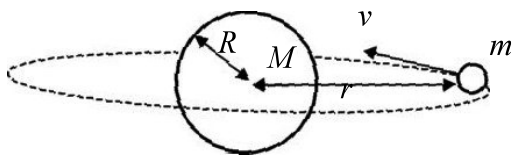


Figure 2.7

A satellite which stays relatively unmoved vertically above a precise point on Earth is called a geo – stationary satellite. The radius of the orbit of a geo – satellite is decided so that its period of revolution is 24 hours.

These satellites which have been launched to employ for communication purposes among various lands on Earth will have to essentially satisfy the following conditions regarding their motion and orbit.

1. Geo – stationary satellite should revolve in the same sense as the sense of rotation of Earth.
2. The angular velocity of revolution of a geo – stationary satellite should be equal to that of the rotation of the Earth. In other words the periodic time of revolution of a geo – stationary satellite should be 24 hours.
3. The orbit of a geo – stationary satellite should be in the same plane as the equator of the Earth.

Considering the periodic time of a satellite as $T = 2\pi\sqrt{\frac{r^3}{GM}}$

Periodic time of a geo – stationary satellite is 24×60×60 = 86400 s.

$$2\pi\sqrt{\frac{r^3}{GM}} = 24 \times 60 \times 60 = 86400\ s$$

$$\therefore \quad 86400 = 2 \times \frac{22}{7}\sqrt{\frac{r^3}{6.7 \times 10^{-11} \times 6.0 \times 10^{24}}}$$

$$r = 42400 \times 10^3\ \text{m}$$

If earth's radius is 6400km,
height of satellite's above the
earth surface = (42400 – 6400) km
= 36000 km

2.3.4 Escape velocity

The minimum velocity with which a body has to be projected from the surface of Earth to over come Earth's gravitation and never return to Earth again is called the escape velocity of Earth.

Considering a body of mass m placed on the surface on Earth of mass M and radius R, its potential energy $E_1 = -G\frac{Mm}{R}$

If this body is projected with a velocity v from the surface of Earth, its total energy,

$$E = \text{Kinetic energy} + \text{Potential energy}$$

$$= \frac{1}{2}mv^2 + \left(-G\frac{Mm}{R}\right)$$

If v is the escape velocity, this total energy will be spent and become zero at an infinite distance. Then according to the law of conservation of energy,

$$\frac{1}{2}mv^2 - G\frac{Mm}{R} = 0$$

$$\frac{1}{2}mv^2 = G\frac{Mm}{R}$$

$$v = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2gr^2}{R}} = \sqrt{2gR}$$

where $G\frac{M}{R^2} = g$ = Gravitational Intensity on the Earth's surface

considering $g = 9.8$ m s^{-2} and the radius of the Earth $R = 6.4 \times 10^6$ m,

Escape velocity of the Earth $= \sqrt{2 \times 9.8 \times 6.4 \times 10^6}$

$= \sqrt{19.6 \times 6.4 \times 10^6}$

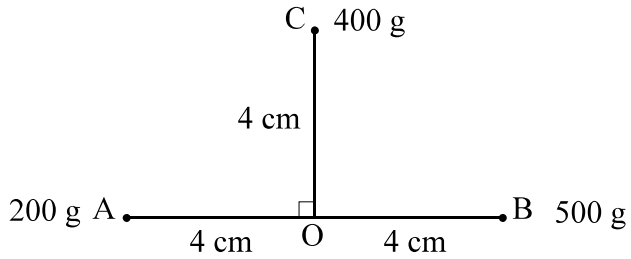
$= \sqrt{196 \times 64 \times 10^4}$

$= 14 \times 8 \times 10^2 = 112 \times 10^2$ m s^{-1}

$= \underline{\underline{11.2 \text{ km s}^{-1}}}$

Solved exercises

1.



At the two ends A and B of a straight line, AB of length 8 cm are placed masses of 200 and 500 g respectively

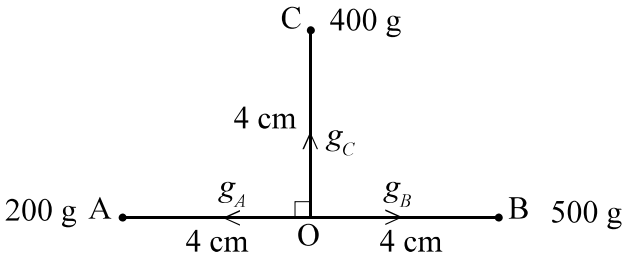
At the end C of the perpendicular bisector, OC of the line AB is placed a mass of 400 the length OC being 4 cm

Calculate,

(a) The resultant gravitational field intensity at the point O.

(b) The total gravitational potential at the point O.

(a)



Considering x Components,

$$\longrightarrow \quad g_1 = g_B - g_A = G\frac{500\times10^{-3}}{(4\times10^{-2})^2} - G\frac{200\times10^{-3}}{(4\times10^{-2})^2}$$

$$= G\frac{300\times10^{-3}}{(4\times10^{-2})^2} = G\frac{3\times10^3}{16}$$

Considering y Components,

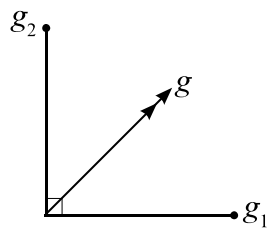
$$\uparrow \quad g_2 = g_C = G\frac{400\times10^{-3}}{(4\times10^{-2})^2} = G\frac{4\times10^3}{16}$$

Gravitational intensity at O,

$$g = \sqrt{g_1^2 + g_2^2}$$

$$= G\frac{10^3}{16}\sqrt{4^2+3^2} = G\frac{5\times10^3}{16}$$

$$= \underline{\underline{3.125\times10^2\,G}}$$



(b) Gravitational potential at O,

$$V = \left[G\frac{200 \times 10^{-3}}{(4 \times 10^{-2})} + G\frac{500 \times 10^{-3}}{(4 \times 10^{-2})} + G\frac{400 \times 10^{-3}}{(4 \times 10^{-2})} \right]$$

$$= \frac{G \times 10^{-1}}{4 \times 10^{-2}}(2 + 5 + 4) = -G\frac{110}{4}$$

$$= \underline{\underline{-27.5\ G\ \text{J kg}^{-1}}}$$